Problem Statement – ATM

In today's fast-paced digital world, banking customers expect 24/7 access to their financial resources without the need to visit a physical branch. An Automated Teller Machine (ATM) bridges this gap by allowing users to carry out essential banking operations such as cash withdrawals, balance inquiries, mini-statements, fund transfers, and PIN changes conveniently and securely.

Despite its widespread use, many ATM systems face limitations such as lack of integration with modern banking APIs, outdated security mechanisms, and non-intuitive user interfaces, leading to poor user experience and increased vulnerability to fraud. There is a growing need for an intelligent and scalable ATM system that not only improves reliability and performance but also enhances security through multi-factor authentication, biometric verification, and real-time fraud detection.

The proposed ATM system aims to modernize traditional ATM operations by supporting:

- Multi-language interfaces for regional accessibility.
- Integration with core banking systems for live account updates.
- Contactless interactions (e.g., NFC, QR-based withdrawals).
- Real-time SMS/email notifications for all transactions.
- Compatibility with chip-based and virtual cards.
- Support for UPI and cardless withdrawals.

The ATM software must also incorporate features such as daily cash limits, account lockout after multiple failed PIN attempts, and end-to-end encryption for data protection. Furthermore, the backend should ensure secure connectivity with the bank's central database, monitor suspicious activities, and log every transaction for auditing purposes.

The primary challenge is to build a secure, efficient, and regulatory-compliant ATM solution that provides uninterrupted banking services while maintaining scalability, fault-tolerance, and ease of maintenance.

Software Requirements Specification – ATM System

1. Introduction

The ATM System is a self-service banking application designed to offer users 24/7 access to essential financial operations, such as cash withdrawals, fund transfers, balance inquiries, and PIN management, without the need for a physical bank visit. The system emphasizes security, reliability, and ease of use, aiming to enhance the overall banking experience.

1.1 Purpose

This SRS document outlines the functional and non-functional requirements for the development of the ATM System. It serves as a reference for:

- Developers – for implementation and testing.
- Testers – for validating requirements.
- Documentation Team – for creating user manuals.
- Management – for project tracking and planning.

1.2 Scope

The ATM System will enable users to:

- Withdraw cash
- Check account balance
- Transfer funds between accounts
- View transaction history
- Change ATM PIN
- Receive SMS/email alerts for transactions

It will integrate with a bank's core system to perform real-time transactions securely. The system will include both software (user interface, backend processing) and hardware interfaces (card reader, keypad, cash dispenser).

Estimated development time: 6 months
Estimated budget: ₹40 lakhs

1.3 Definitions, Acronyms, and Abbreviations

- ATM: Automated Teller Machine
- PIN: Personal Identification Number
- NFC: Near Field Communication
- OTP: One-Time Password
- API: Application Programming Interface
- AES: Advanced Encryption Standard
- KYC: Know Your Customer
- DBMS: Database Management System

1.4 References

- RBI Guidelines for ATM Transactions
- ISO 9564 (PIN Security Standard)
- Core Banking API Documentation

1.5 Overview

- Section 2: General description of the system
- Section 3: Functional and non-functional system features
- Section 4: External interfaces and communications

2. Overall Description

2.1 Product Perspective

The ATM System is part of a larger banking infrastructure that communicates with core banking servers to perform secure and real-time transactions. It comprises:

- User Interface: Touchscreen/keypad interface for user interaction
- Hardware Modules: Card reader, cash dispenser, receipt printer, biometric scanner (optional)
- Backend System: Core banking API integration and real-time processing engine
- Security Layer: Manages encryption, fraud detection, and user authentication

2.2 Product Functions

- Authentication: Login via card + PIN and optional OTP or biometric verification
- Cash Withdrawal: Choose amount and receive cash with receipt
- Balance Inquiry: View real-time account balance

- Mini Statement: View last 5–10 transactions
- Fund Transfer: Transfer money between user's accounts or third-party accounts
- PIN Change: Change ATM PIN securely
- Transaction Alerts: Send SMS/email confirmation for every action

2.3 User Characteristics

- General Users: Must be able to perform basic banking actions without prior training
- Elderly Users: Simplified UI with large text and multi-language support
- Administrators: For system maintenance, monitoring cash levels, and error logs

2.4 Constraints

- Must support 50,000 concurrent sessions
- Cash withdrawal limited to ₹10,000 per transaction, ₹50,000 per day
- Transactions must complete within 10 seconds
- Must follow PCI-DSS, ISO 9564, and RBI security guidelines
- Encrypted storage and transmission using AES-256

2.5 Assumptions and Dependencies

- Internet connection is available
- Integration with bank's core system APIs
- Power supply with UPS/failover support
- Availability of biometric scanner (if used)

3. System Features

3.1 Functional Requirements

Authentication

- Card and PIN required for access
- Optional OTP or biometric login
- Lock account after 3 invalid PIN attempts

Cash Withdrawal

- Withdraw predefined or custom amounts
- Show available balance
- Issue printed/e-receipt

Balance Inquiry

- Display current and available balance
- Show masked account number

Fund Transfer

- Transfer to self or registered third-party accounts
- Enter recipient account, IFSC, and amount

Mini Statement

- Display last 5–10 transactions
- Option to print or email

PIN Management

- Change PIN after identity verification
- Use OTP or existing PIN for confirmation

Transaction Alerts

- Notify user via SMS or email
- Alerts for success/failure, transfers, withdrawals

4. External Interface Requirements

4.1.1 User Interface Requirements

- Multi-language support
- Touchscreen or physical keypad interface
- Audio instructions for visually impaired users

4.1.2 Hardware Interface Requirements

- Card reader (EMV chip + magnetic strip)
- Cash dispenser with tray sensor
- Receipt printer
- Biometric scanner (optional)
- UPS for power backup

4.1.3 Software Interface Requirements

- Bank’s core system APIs
- Notification service (SMS/Email APIs)

- Security libraries for AES encryption
- Local DBMS (e.g., SQLite) for temporary caching

4.1.4 Communication Interface Requirements

- HTTPS connection to bank servers
- Encrypted communication via SSL/TLS
- Redundant connectivity for failover

5. Non-Functional Requirements

5.1 Performance Requirements

- Handle up to 50,000 users concurrently
- Response time: ≤ 10 seconds per transaction
- Daily transaction volume: ₹1 crore across the network

5.2 Software System Attributes

- Reliability: Uptime ≥ 99.99%
- Security: AES-256 encryption, 2FA, secure PIN handling
- Scalability: Easily add more ATM units
- Maintainability: Remote diagnostics and alerts
- Usability: Intuitive UI, audio prompts, and accessibility features